

ENGINEERING TECHNICAL REFERENCE MANUAL & SPECIFICATION

SiWG917 Dual-Node V2X Telemetry Prototype Platform

Cooperative Telemetry Datagram Protocol, IMU Ingestion Engine & FreeRTOS Dual-Tasking

Target SoC: Silicon Labs SIWG917M111MGTBA	Radio Board: BRD2605A (Rev A02)	SDK Base: Simplicity SDK 2025.12.3 / WiSeConnect 3
Wire Protocol: v2x_frame_t (42 Octets Packed)	Binary Format: .rps (RAM Execution Image)	Toolchain: GNU ARM Embedded Toolchain (GCC 12.2.rell)

1. EXECUTIVE PLATFORM ARCHITECTURE

The SiWG917 V2X Telemetry Node platform is an experimental prototype designed for cooperative vehicular awareness, physical state broadcast, and inertial dynamic telemetry. The architecture decouples high-bandwidth 802.11 b/g/n wireless datagram transmission from deterministic SPI sensor acquisition across the dual-core architecture of the Silicon Labs SiWG917. The ARM Cortex-M4 application core coordinates FreeRTOS execution, frame validation, and ULP sensor drivers, while the Network Wireless Processor (NWP) co-core executes the 802.11 MAC, WPA2 supplicant, and embedded TCP/IP stack.

2. MICROCONTROLLER CORE & HARDWARE CLOCK DISTRIBUTION

Peripherals are distributed across isolated power states to optimize peripheral consumption:

Subsystem	Power Domain	Clock Source & Routing	Operating Parameter / Mode	Hardware Driver
Application Processor	Base MCU (M4 Core)	M4 PLL / High-Frequency RC	180 MHz active CPU clock	system_si91x.c
ULP SSI Master	ULP Domain (Power State 2)	ULPCLK_GPIO / Primary Divider	10.0 MHz SPI clock, CPOL=0 / CPHA=0	sl_si91x_ssi.c
Sensor Rail Switch	UULP (Ultra-ULP) NPSS	UULP_VBAT_GPIO logic core	NPSS GPIO Pin 1, MUX Mode 1 (Push-Pull)	sl_si91x_driver_gpio.c
Baseband NWP Co-Core	NWP Subsystem	RF Synthesizer / XTAL	2.4 GHz Band, 802.11b/g/n (Auto/Ch 6)	sl_wifi.c / sl_net.c
System Millisecond Timer	SYSRTC Sleep Timer	32.768 kHz Low-Power Crystal / RO	Monotonic uptime & tick conversion	sl_sleeptimer.c

3. MOTION SENSOR POWER SEQUENCING & ULP SSI CONFIGURATION

The InvenSense ICM40627 6-axis IMU resides on a switched NPSS power plane to prevent parasitic diode back-powering during inactive intervals. Cold boot enforces strict timing settling delays:

- Clock Gate & Rail Assert:** Enables `ULPCLK_GPIO`, assigns NPSS GPIO Pin 1 to Mux Mode 1 (Push-Pull Output), and drives the pin HIGH.
- Decoupling Capacitor Stabilization:** Enforces a mandatory 25 ms delay via SYSRTC before initiating bus transactions.
- Bus Settling & Software Reset:** Initializes the SSI peripheral, waits 10 ms, issues a software register reset, and allows a 15 ms stabilization window.

Hardware Electrical Invariant: Never assert SPI transactions to the ICM40627 while NPSS GPIO 1 is unasserted (low). Parasitic conduction through the SPI MISO/MOSI ESD clamp diodes into the unpowered sensor plane can cause peripheral latch-up and drop the ULP power rail.

ULP SSI Interface Parameters

- Target Peripheral:** `SL_SSI_ULP_PRIMARY_ACTIVE` configured in `SL_SSI_ULP_MASTER_ACTIVE` at 10.0 MHz.
- Serial Framing:** 8-bit frame transmissions, Clock Polarity `CPOL=0` (idle low), Clock Phase `CPHA=0` (sample on leading edge).
- Hardware Chip Select:** Routed to `SSI_SLAVE_0`.
- Silicon Verification:** Reads register `WHO_AM_I` (address `0x75`) and confirms signature value `0x4E`.

4. IMU SENSOR ENGINE & RING BUFFER REALITIES

The sensor driver in `accelerometer.c` provides hardware abstraction and circular sample buffering for kinematics ingestion:

Buffer Symbol / Macro	Memory Allocation	Nominal Profile	Observed Execution Characteristics
<code>s_sample_buffer[]</code>	1,200 Bytes (SRAM)	100 Elements	Continuous circular ring buffer storing IEEE 754 float vectors (X, Y, Z in g).
<code>IMU_BUFFER_CAPACITY</code>	400 Bytes index	100 Samples	Nominally 4.0 s at 25 Hz; actual span is ~12.6 s under the current ~8 Hz transmit-coupled polling loop.
<code>s_head_index</code>	4 Bytes (volatile)	Pointer	Index pointing to the next writable slot in the circular buffer.
<code>s_total_samples</code>	4 Bytes (volatile)	Counter	Monotonically incrementing sample count since driver initialization.
<code>ACCELEROMETER_FREQ</code>	N/A (Header define)	25 Hz (40 ms)	Driver ODR target. True execution cadence is locked to the <code>v2x_net_task</code> transmission loop (~8 Hz).
<code>SEQUENCE_LENGTH</code>	N/A (Header define)	90 Samples	Nominally 3.6 s history; under ~8 Hz sampling, window expands to ~11.3 s of continuous motion history.

Ring Buffer Mechanics & Decoupled Access

- **Synchronous Ingestion** (`accelerometer_poll`): Invoked by the network task immediately prior to frame packaging. Queries ICM40627 acceleration registers over the SSI bus, scales raw integer counts to engineering units (g), copies the vector into `s_sample_buffer[s_head_index]`, advances the head index modulo 100, and increments `s_total_samples`.
- **Instantaneous Latch Access**: The helper function `accelerometer_get_latest()` computes $(s_head_index - 1 + CAPACITY) \% CAPACITY$ to return the most recently converted physical reading in $O(1)$ time without bus contention or thread locks.
- **Historical Sequence Retrieval**: `accelerometer_read()` unrolls circular buffer indices backwards from the current head to provide a contiguous memory block of N previous samples.

5. COORDINATE SYSTEM & STATIC MOUNTING ORIENTATION

Physical acceleration vectors represent vehicle-relative body coordinates, but require explicit board mounting constraints:

- **Nominal Reference Frame**: Positive X points along vehicle longitudinal forward travel; positive Y points laterally to the right; positive Z points normal to the ground plane upward. In an ideal, flat, level horizontal resting mount, gravity registers as $Z \approx +1.0\text{ g}$ ($X \approx 0.0\text{ g}$, $Y \approx 0.0\text{ g}$).
- **Observed Testbench Vector** ($X \approx 0.97\text{ g}$, $Y \approx 0.08\text{ g}$, $Z \approx 0.21\text{ g}$): Observed baseline readings indicate the development kit was mounted vertically on its edge or substantially pitched during testing.
- **Static Calibration Requirement**: Acceleration thresholds cannot reliably trigger on raw acceleration alone without boot-time gravity vector subtraction or strict chassis alignment. A stationary zero-velocity calibration offset or full IMU rotation matrix must be evaluated prior to active threshold monitoring.
- **Dynamic Dynamic Range**: ICM40627 is configured for $\pm 4\text{ g}$ or $\pm 8\text{ g}$ full-scale range, providing sufficient dynamic headroom for vehicle braking and cornering maneuvers while maintaining sub-millig resolution.

6. WIRE PROTOCOL SPECIFICATION: BASIC SAFETY MESSAGE ('V2X_FRAME_T')

To prevent compiler padding anomalies across architectures, all wireless frames enforce strict 1-byte packing (`#pragma pack(push, 1)`). The over-the-air payload footprint is exactly 42 octets:

Offset (Hex)	Field Identifier	Data Type	Octets	Wire Encoding & Behavioral Protocol Constraints
0x00 - 0x03	magic	uint32_t	4 B	Value 0x56325831 ("V2X1"). Transmitted on wire in little-endian byte order: 0x31, 0x58, 0x32, 0x56.
0x04	version	uint8_t	1 B	Protocol schema revision. Evaluated as 0x01 by current firmware build.
0x05	message_type	uint8_t	1 B	Frame semantic identifier. Current implementation transmits 0x01 (V2X_MSG_TYPE_PERIODIC).
0x06 - 0x07	payload_length	uint16_t	2 B	Total length of wire frame: 42 Bytes (<code>sizeof(v2x_frame_t)</code>).
0x08 - 0x09	sequence	uint16_t	2 B	Monotonically incrementing packet counter for datagram loss and jitter tracking.
0x0A - 0x0F	sender_id[6]	uint8_t[6]	6 B	Hardware IEEE 802.11 MAC address of transmitting interface (retrieved via <code>sl_wifi_get_mac_address</code>).
0x10 - 0x13	timestamp_ms	uint32_t	4 B	Local node uptime in milliseconds converted from SYSRTC sleep timer ticks.
0x14 - 0x17	accel_x	float (IEEE 754)	4 B	Longitudinal acceleration in g units. Aligned on 4-byte boundary.
0x18 - 0x1B	accel_y	float (IEEE 754)	4 B	Lateral acceleration in g units. Aligned on 4-byte boundary.
0x1C - 0x1F	accel_z	float (IEEE 754)	4 B	Vertical acceleration in g units. Aligned on 4-byte boundary.
0x20	gesture	uint8_t	1 B	Kinematic event ID placeholder. Shipped builds transmit 0x03 (NO_GESTURE).
0x21	emergency	uint8_t	1 B	Safety flag. Shipped builds transmit 0x00 pending runtime hysteresis activation.
0x22 - 0x23	reserved	uint16_t	2 B	Alignment padding word to maintain 4-byte offset for confidence. Zeroed during <code>v2x_seal_frame</code> .
0x24 - 0x27	confidence	float (IEEE 754)	4 B	Confidence metric (0.000 to 1.000). Transmitted as 0.0f in current telemetry release.
0x28 - 0x29	checksum	uint16_t	2 B	Fletcher-16 checksum evaluated across preceding 40 payload octets (offsets 0x00 to 0x27).

7. FLETCHER-16 CHECKSUM ALGORITHM

The frame seal algorithm evaluates two independent modular accumulators across the 40 octets preceding the checksum:

```
Let B = {b_0, b_1, b_2, ..., b_{N-1}} where N = offsetof(v2x_frame_t, checksum) = 40 octets.
Initialize: S_1 = 0, S_2 = 0
For k = 0 to N-1:
    S_1 = (S_1 + B[k]) mod 255
    S_2 = (S_2 + S_1) mod 255
Result: checksum = (S_2 << 8) | S_1
```

8. FIVE-STAGE INBOUND FRAME VALIDATION PIPELINE

The receiving node processes incoming datagrams through a strict five-stage gate in `v2x_process_incoming_peer_frames()`:

- 1. **Byte Length Gate:** Asserts received payload size exactly equals `sizeof(v2x_frame_t)` (42 Bytes) to reject fragments.
- 2. **Magic Signature Gate:** Asserts `frame->magic == 0x56325831` to drop unintended UDP port 5000 traffic.
- 3. **Schema Version Gate:** Asserts `frame->version == 1` to prevent processing incompatible frame layouts.
- 4. **Checksum Verification Gate:** Invokes `v2x_validate_frame()`, zeroing the checksum field in a local copy and evaluating whether the recomputed Fletcher-16 word matches the received value.
- 5. **Local MAC Echo Rejection:** Evaluates `memcmp(frame->sender_id, s_local_mac, 6)` to silently drop reflected self-broadcasts.

9. FREERTOS DUAL-TASK ARCHITECTURE & EXECUTION MODEL

Networking is partitioned into two decoupled FreeRTOS tasks to isolate periodic transmission from blocking socket calls:

Task Symbol	Priority Level	Stack Allocation	Operational Role & Execution Dynamics
v2x_net_task_entry	osPriorityNormal	3,584 Bytes	Initializes Wi-Fi and IMU, samples sensor, packages frames, applies jitter, and issues <code>sendto()</code> .
v2x_rx_task_entry	osPriorityBelowNormal	3,072 Bytes	Runs dedicated blocking <code>recvfrom()</code> loop, validates incoming frames, and logs peer metrics.

10. TRANSMISSION JITTER & EFFECTIVE TELEMETRY CADENCE

To mitigate persistent RF packet collisions on 802.11 CSMA/CA networks, transmission timing incorporates an xorshift-based non-deterministic delay:

```
static uint32_t v2x_get_tx_jitter_ms(void) {
    uint32_t tick = sl_sleeptimer_get_tick_count();
    uint32_t value = tick ^ (tick >> 7) ^ (tick << 9);
    return value % (V2X_TX_JITTER_MAX_MS + 1U); /* Range: 0 to 20 ms */
}
```

Timing Analysis: The scheduling loop invokes `osDelay(jitter_ms)` within the active deadline and computes the next deadline via `now + V2X_TX_PERIOD_MS`. Because the 100 ms baseline interval accumulates the 0–20 ms delay plus sensor SPI and socket overhead, real-world frame cadences average approximately 126 ms (~8 Hz effective broadcast rate).

11. NWP SOCKET INTEGRATION & PROTOCOL QUIRKS

- **Broadcast Socket Option (`SO_BROADCAST`):** Executing `setsockopt(..., SO_BROADCAST, ...)` returns `errno = 42`. In the toolchain socket layer this indicates unsupported socket operation; however, the NWP offload co-processor natively forwards broadcast datagrams without explicit option configuration. The driver logs a warning and proceeds normally.
- **Subnet vs Global Broadcast Routing:** The Soft-AP node broadcasts to the global broadcast address (`255.255.255.255:5000`), whereas Station nodes target directed subnet broadcast (`192.168.10.255:5000`).
- **Blocking Ingress Isolation:** Because non-blocking flags (`O_NONBLOCK`) and receive timeouts (`SO_RCVTIMEO`) are unsupported by the underlying NWP socket wrapper, RX processing must remain isolated in `v2x_rx_task_entry` to prevent starvation of the telemetry scheduling thread.
- **Stack Overflow Monitor:** Implements `vApplicationStackOverflowHook()`. If task memory breaches its watermark, the hook traps execution, logs the offending task name over VCOM, and enters a `while(1)` barrier.

12. KINEMATIC DECELERATION HYSTERESIS ARCHITECTURE

The kinematic deceleration classifier is designed to isolate emergency panic braking from chassis vibration using a dual-counter Finite State Machine (FSM). *Note: This subsystem represents the reference algorithm designed for integration into future builds:*

FSM Control Parameter	Reference Constant	Calibrated Value	Functional Behavior & Vibration Suppression Mechanism
Monitored Vector Axis	BSM_LONGITUDINAL_AXIS	Axis 0 (Longitudinal X)	Monitors longitudinal acceleration along vehicle direction of travel.
Emergency Threshold	BSM_DECEL_THRESHOLD_G	-0.50 g	Flags deceleration exceeding nominal service braking (typical panic stop > 0.45 g).
Confirmation Counter	BSM_DECEL_CONFIRM_SAMPLES	3 Consecutive Samples	Requires 3 consecutive samples below threshold (~380 ms at 8 Hz) to declare Emergency.
Release Recovery Counter	BSM_DECEL_RELEASE_SAMPLES	5 Consecutive Samples	Requires 5 consecutive samples above threshold (~630 ms at 8 Hz) to return to Safe state.

13. FINITE STATE MACHINE TRANSITIONS & LATCH INTEGRITY

The state machine coordinates transitions across three discrete operating modes:

- BSM_STATE_SAFE (Nominal):** Steady-state driving ($a_x > -0.50\text{ g}$). Confirmation counter held at 0; emergency flag set to false.
- BSM_STATE_DECEL_WARNING (Debouncing):** 1 or 2 samples drop $\leq -0.50\text{ g}$. Confirmation counter advances; over-the-air emergency alert remains unasserted to suppress transient false alarms.
- BSM_STATE_EMERGENCY (Latching Alert):** Confirmation count reaches 3. Sets `emergency = true` and asserts `BSM_EVENT_SUDDEN_DECELERATION`.

14. HARDENED FSM ALGORITHMIC LOGIC

To prevent signal chatter when operating near the -0.50 g boundary, the emergency state must remain latched while debouncing potential recovery:

```
void bsm_update_hardened(float accel_x_g) {
    s_longitudinal_accel_g = accel_x_g;

    if (s_longitudinal_accel_g <= BSM_DECEL_THRESHOLD_G) {
        s_release_count = 0U;
        if (s_confirm_count < BSM_DECEL_CONFIRM_SAMPLES) {
            s_confirm_count++;
        }
        /* Latch emergency state: once asserted, do not clear on transient drops */
        if (s_confirm_count >= BSM_DECEL_CONFIRM_SAMPLES || s_emergency) {
            s_state      = BSM_STATE_EMERGENCY;
            s_event      = BSM_EVENT_SUDDEN_DECELERATION;
            s_emergency = true;
        } else {
            s_state      = BSM_STATE_DECEL_WARNING;
            s_event      = BSM_EVENT_NONE;
            s_emergency = false;
        }
    } else {
        s_confirm_count = 0U;
        if (s_emergency) {
            if (++s_release_count >= BSM_DECEL_RELEASE_SAMPLES) {
                s_state      = BSM_STATE_SAFE;
                s_event      = BSM_EVENT_NONE;
                s_emergency = false;
                s_release_count = 0U;
            }
        } else {
            s_state      = BSM_STATE_SAFE;
            s_event      = BSM_EVENT_NONE;
            s_emergency = false;
        }
    }
}
```

15. VERIFICATION TRACES (ABRIDGED CONSOLE CAPTURE)

The following console output traces capture bidirectional frame exchange between Access Point Node A and Station Node B operating at 115200 baud, 8N1 (abridged for brevity):

Node A Console (Access Point Master - COM11)

```
=====
                        SiWG917 V2X PLATFORM
=====

[APP] Creating network task...
[V2X NET] Initializing Network Subsystem (Soft AP)...
[V2X NET] AP Active: SSID='V2X_AP_NODE', IP=192.168.10.1
[V2X NET] MAC = 08:DD:EB:CF:46:58
[IMU] ICM40627 ID 0x4E verified.
[IMU] Initialized at 25 Hz sampling rate.
[V2X NET] WARNING: SO_BROADCAST failed, errno=42. Continuing.
[V2X NET] UDP socket ready on port 5000.
[V2X RX] RX TASK STARTED (Buffer: 42 bytes)
[V2X TX] OK | Seq=0 | Bytes=42 | Accel=(0.971, 0.072, 0.211) g | Peer=NO
... [steady-state execution] ...
[V2X TX] OK | Seq=4083 | Bytes=42 | Accel=(0.971, 0.072, 0.211) g | Peer=YES
[V2X TX] OK | Seq=4084 | Bytes=42 | Accel=(0.965, 0.089, 0.245) g | Peer=YES
```

Node B Console (Station Client - COM21)

```
=====
                        SiWG917 V2X STATION NODE + ICM40627
=====

[V2X NET] Scanning 2.4 GHz channels for AP 'V2X_AP_NODE'...
[V2X NET] Station interface is UP.
[V2X NET] Station MAC = 08:DD:EB:CF:3F:A0
[IMU] ICM40627 ready on Board B.
[V2X NET] UDP socket bound to port 5000.
[V2X NET] Broadcast destination: 192.168.10.255:5000
[V2X RX] RX TASK STARTED
[V2X RX] recvfrom returned: 42 bytes
[V2X RX] VALID PEER PACKET | Seq=4086 | MAC=08:DD:EB:CF:46:58 | Accel X=0.984 g | Checksum=0x6100
```

16. FIRMWARE ARCHITECTURE & SOURCE MANIFEST

File Path	Subsystem Layer	Primary Exported Symbols & Implementation Responsibilities
app_soft_ap.c	Access Point App	Soft AP initialization, periodic broadcast scheduling, dedicated blocking UDP RX task.
app_station.c	Station App	Wi-Fi client association, DHCP IP acquisition, subnet-directed UDP broadcast telemetry.
accelerometer.c / .h	Sensor Ingestion	NPSS GPIO 1 power gating, 10 MHz ULP SSI master driver, 100-element circular buffer.
bsm_packet.h	Wire Protocol	42-byte packed v2x_frame_t definition, Fletcher-16 seal and validation logic.
constants.h / app.h	Configuration	UDP port definitions, network credentials, timing macros, and thread definitions.
rgb_led.c / .h	Board Diagnostics	Hardware tri-color LED driver indicating network join status and error traps.

17. HARDWARE DEPLOYMENT & FLASHING RUNBOOK (.RPS PROFILE)

The SiWG917 packages execution binaries with boot headers and RAM metadata into .rps images via v2_x.s1pb. Flashing is executed via Simplicity Commander:

```
# 1. Enumerate connected SiWG917 radio boards
$ commander adapter list

# 2. Flash generated RPS image to Access Point (Board A)
$ commander rps load v2_X_ap.rps --device SIWG917M111MGTBA --serialno 440332563

# 3. Flash generated RPS image to Station Node (Board B)
$ commander rps load v2_X_sta.rps --device SIWG917M111MGTBA --serialno 440332449
```